MILITARY SPECIFICATION
INDICATOR, INDICATED AIRSPEED, 20-250 KNOTS

This specification is mandatory for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers pitot static airspeed indicators, range 20 to 250 knots.

*1.2 Classification. The indicators shall be furnished in the following part numbers:

MS28021-2	Lusterless White Dial
MS28021-3	Lusterless White Dial
MS28021-4	Lusterless White Dial

2. APPLICABLE DOCUMENTS

*2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-I-5415	Indicators, Airspeed, Pitot and Static Pressure, General Specification for

Federal

PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-636	Box, Fiber Board

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-794	Parts and Equipment, Procedure for Packaging and Packing of
MIL-STD-831	Test Reports, Preparation of

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MS28021	Indicator - Indicated Airspeed, Pitot Static, 20-250 Knots
MS33585	Pointer, Dial, Standard Design of Aircraft Instrument

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The indicator furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 General. The general requirements specified in MIL-I-5415 are applicable as general requirements of this specification. Where the requirements of MIL-I-5415 and this specification conflict, the requirements of this specification shall govern.

3.3 Material. Material shall be in accordance with the requirements specified in MIL-I-5415.

3.4 Design and construction.

3.4.1 Range. The airspeed indicator shall be designed to satisfactorily operate in the range of 20 to 250 knots.

3.4.2 Dimensions. The airspeed indicator shall conform to the dimensions shown on MS28021.

3.4.2.1 Case finish. The case shall be finished with a lusterless black material, color 37038 of FED-STD-595. The finishing material shall be of a durable type to withstand usage encountered in service.

*3.4.3 Pointer. The pointer shall conform to MS33585-3. The shaded portion of the pointer shall be finished in lusterless white, color 37857 of FED-STD-595. The unshaded portion of the pointer shall be finished in lusterless black, color 37038 of FED-STD-595.

3.4.3.1 Limitation of pointer movement. The pointer movement shall be limited by suitable stops in the mechanism in such a way that the pointer will not be permitted to rotate more than 355° of an arc clockwise from the zero graduation, when the indicator is submitted to overpressure tests. Suitable stops may also be incorporated in the indicator mechanism to limit the counterclockwise rotation of the pointer to the zero graduation. The design of the stops shall be subject to the approval of the procuring activity.

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3.5 Weight. The weight of each indocator shall not exceed 1 pound.

3.6 Performance. The airspeed indicators shall perform satisfactorily when subjected to the tests specified in section 4.

3.7 Markings.

3.7.1 Dial. Dial markings shall be durable, and the size, material, and finish shall conform to Figure 1.

3.7.2 Visibility of dial markings. All graduations, numerals, pointers, and other markings of the dial shall be readable from any point within the frustum of a cone, the sides of which makes an angle of not less than 30° with a perpendicular to the dial and the small diameter of which is the aperature of the dial.

*3.8 Identification of product. The indicator shall be marked in accordance with MIL-STD-130.

3.9 Workmanship. Workmanship shall be in accordance with MIL-I-5415.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. Qualification inspection

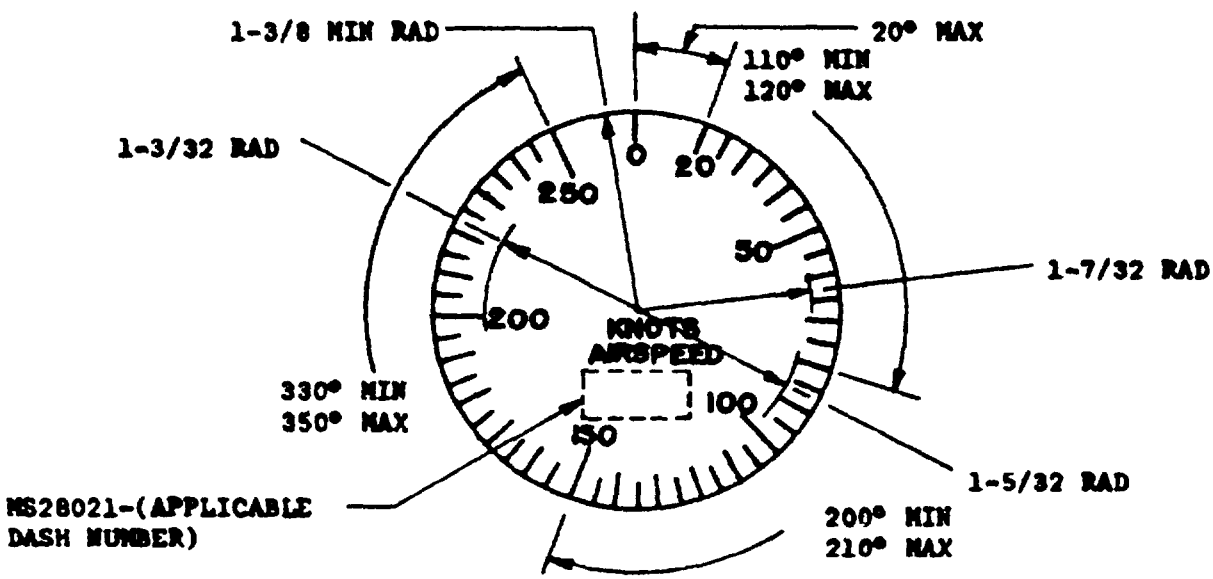
2. Quality conformance inspection

4.2.1 Qualification inspection.

*4.2.1.1 Qualification test samples. The test samples shall consist of 3 indicators which are respresentative of production items.

*4.2.1.2 Test report and test samples. When the tests are conducted at a location other than the laboratory of the qualifying activity, three copies of a test report in accordance with MIL-STD-831, and the samples which were tested shall be furnished.

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ALL ARC DIMENSIONS START AT ZERO POSITION.
ALL GRADUATIONS ARE PARTIAL RADIAL LINES.
DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: FRACTIONS ± 1/64.

-Markings	Height or Length ± .010	Width of Line or Graduation ± .005	Material or Finish
Graduations corresponding to numerals 50, 100, 150, 200 and 250	As shown on figure	.031	Lusterless White
Numerals 50, 100, 150, 200, 250	.312	.031	
Graduations other than those specified	As shown on figure	.016	
Numerals 0 and 20	.156	.016	
Graduation Corresponding to 0	.125	.016	
Lettering "KNOTS"	.188	.016	
Lettering "MS28021-	.047	.016	
Lettering "AIRSPEED"	.047	.016	Lusterless Black
Background of dial			

FIGURE 1. Dial Markings

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4.2.1.3 Qualification test. The sample indicators shall be subjected to all the tests described under test methods in MIL-I-5415 except vibration tests. The vibration failure and vibration error tests shall be conducted as described in 4.3.1 and 4.3.2 of this specification. The test values of tables I and II, specified herein, shall be used for the qualification test.

*4.2.2 Quality conformance inspection. The quality conformance inspection shall be conducted as specified under Quality Conformance tests in MIL-I-5415. The vibration tests shall be conducted as specified in 4.4.1 and 4.4.2 of this specification. The test values of tables I and II, specified herein, shall be used for the quality conformance tests.

4.3 Test conditions. Unless otherwise specified, the test conditions shall be as specified in MIL-I-5415.

4.4 Test methods. The test methods shall be as specified in MIL-I-5415 except that the vibration test methods shall be as follows.

4.4.1 Vibration failure. This test shall be conducted using the test differential pressure specified in table I as the test point. The instrument indication at the test differential pressure specified shall be determined with pressures increasing. The indicator shall be subjected to vibration with a circle diameter between 0.018 and 0.020 of an inch for a 1-hour period at a frequency of 1,000 cycles per minute, followed by a 1-hour period at a frequency of 2,000 cycles per minute and then followed by a 1-hour period at a frequency of 3,000 cycles per minute. At the end of the vibration period, the indicator shall be subjected to the pressure specified to produce the test point reading (with pressure increasing). The difference between this indication and that determined prior to the vibration for the same differential pressure shall not exceed the values specified in table I. No looseness in the mechanism or damage to any part of the indicator shall result from this test.

4.4.2 Vibration Error. The indicator shall be mounted on a vibration stand with a line joining the centers of the two lower mounting holes in the horizontal plane, with the plane of the mounting lugs of the indicator vertical. This test shall be conducted while the differential pressures specified in table I are applied to the indicator. The indicator shall be subjected to vibration with a circle diameter between 0.003 and 0.005 of an inch at frequencies from 500 to 3,000 cycles per minute, to determine that the natural frequency of the indicator does not occur in this frequency range. While the indicator is being vibrated, the maximum total spread of pointer vibration and the pointer variation from its original position shall not exceed the values specified in table I.

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TABLE I
Test Values

Tests	Tolerances (Values in knots unless otherwise specified)
DAMPING	
Time required from full scale deflection to 20 knots	1.0 $^{+0.5}_{-0.2}$ second
LOW TEMPERATURE EXPOSURE AND OPERATION	2.5
HIGH TEMPERATURE OPERATION	2.0
AGING	2.0
VIBRATION FAILURE	
Test differential pressure equivalent to 170 knots	2.5
FRICTION	3.0 at 20 knots Test Point 2.0 at all other Test Points
POSITION ERROR	2.0
POINTER ZERO POSITION	
Indication at 0	± 1/16 inch
VIBRATION ERROR	
Test differential pressure equivalent to 40 and 200 knots	2.0
OVERPRESSURE	
Pitot pressure connection - 75 inches of water	2.0
Static pressure connection - 6.0 inches of water	2.0
SEASONING	2.0

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TABLE II
Scale Error at Room Temperature

Speed (Knots)	Pressure (Inches of Water)	Allowable Scale Errors (Knots)
20*	0.2611	3.5
30	0.5876	3.5
40*	1.045	3.5
50	1.634	2.0
60*	2.354	2.0
70	3.207	2.0
80*	4.192	2.0
90	5.310	2.5
100*	6.563	2.5
110	7.951	2.5
120*	9.475	2.5
130	11.14	2.5
140*	12.94	3.0
150	14.87	3.0
160*	16.95	3.0
170	19.17	3.0
180*	21.54	3.0
190	24.05	3.0
200*	26.71	3.5
210	29.51	3.5
220*	32.47	3.5
230	35.58	3.5

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TABLE II
(Cont'd)

Speed (Knots)	Pressure (Inches of Water)	Allowable Scale Errors (Knots)
240*	38.84	3.5
250	42.27	3.5

*Test points

*4.5 Inspection of the preservation, packaging, packing and marking for shipment and storage shall be conducted in accordance with the Quality Assurance Provisions of MIL-STD-794 to determine conformance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

*5.1.1 Packaging for each indicator shall be level A or C of MIL-STD-794 as specified in the contract. Indicators packaged Level A shall be in accordance with preservation method IIC of MIL-P-116. The unit container shall conform to PPP-B-636, weather resistant class.

*5.2 Packing

*5.2.1 Level A.- Level A packing shall be in accordance with MIL-STD-794. Containers shall conform to PPP-B-636, weather resistant boxes, with exceptions specified in 5.2.1.1.

*5.2.1.1 Containers conforming to PPP-B-601 may be used when necessary to meet minimum departmental requirements.

*5.2.2 Level B. Level B packing shall be in accordance with MIL-STD-794. The container shall conform to PPP-B-636, domestic class.

*5.2.3 Level C. Level C packing shall be in accordance with level C packing of MIL-STD-794.

*5.3 Marking and identification. Unit packages and shipping containers shall be marked and identified in accordance with MIL-STD-129.

6. NOTES

*6.1 Intended use. The airspeed indicators are intended for use on aircraft to indicate the speed relative to the air at sea level and, when at any other altitude, the equivalent sea level speed corresponding to the actual sustaining force of the air.

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6.2 The notes of MIL-I-5415 are applicable to this specification.

6.3 Certain provisions (case configuration and location of the zero position on the dial) of this specification are the subject of international standardization agreement (ABC 17/20 and STANAG 3434). When amendment, revision, or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

*NOTE: The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians
Army - AV
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Air Force - 71

Preparing Activity
Air Force - 71

Project Number:
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side, staple in corner, and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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